Eng Int Syst (2011) 4: 1–7
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Engineering
Intelligent Systems

A novel model for measuring B2B transaction using intelligent agent and fuzzy system

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Measuring the interoperability is a common tool to assess the effectiveness of B2B. This model is introducing a common infrastructure to facilitate the B2B interoperability measurement. It provides a repository for the businesses and forces them to register. Later, the registered businesses will get the required intelligent agents which are in charge of measuring the interoperability. They use ebXML and web services in B2B Platform in their company to accomplish their tasks. Intelligent agents collect all permitted data by negotiating with the repository and other intelligent agents such as financial interactions and trading volume .The collected data by agents could be used to optimize the quality and volume of interactions between different businesses.

Keywords: e-Commerce, ebXML , Intelligent Agents, B2B, interoperability

1. INTRODUCTION

Nowadays, the modern economy is impacted heavily by the "digital economy" due to its significant new advantages. However, still there are some areas that are not providing the desired level of services to the public such as good data collection tools. Lack of enough public statistical data pertaining to the B2B financial transactions has turned out to be a real challenge for policy makers to recognize political priorities, allocating resources, designing, evaluating and revising business guidelines and electronic business. In fact, since the methods of data collection for electronic business are both ineffective and costly, it has been decided to collect this data on long-term plans. Thus, the businesses and investigators do not have access to updated data to make right decisions. Traditional methods such as face to face or self-report are used to collect data in the existing methods of measuring the bulk of electronic business which is time-consuming, more prone to mistakes, and of course more costly. Moreover, the continuous changes in electronic business technologies and also using modern methods in doing business entail changes in the data collection methods. At the present method, this change may take months or years to be included in the questionnaire.

2. BACKGROUND

According to the statistics revealed by OECD (Organisation for Economic Co-operation and Development) in 2009 [1],

the bulk of financial transactions done in B2B equals 95% of total business in the world, this amount for B2C equals 5% . Research shows that the amount of business done electronically inside a country is more than foreign electronic business of that country. As it was shown, the amount of B2B in internal electronic business plays an important role in the financial business of studied countries.

As discussed, measurement is very critical and proposing measurement scheme for electronic commerce is not a straightforward task for a number of reasons; including defining what constitutes electronic commerce, the speed of its growth and evolution, and the fact that in many cases firms conduct both electronic commerce and traditional commerce simultaneously. Moreover, according to [1] "quantifying the values associated with electronic commerce activities can be challenging since many of its key qualities – convenience, variety, and ease of access to information – are difficult to be measured". This leads to a situation where the traditional methods of data collection is not working anymore and new measurement/data collection methods needs to be implemented for electronic commerce [2]. It should be noted that the scope of this work is only on the traditional ecommerce activities and does not include m-commerce, etc.

3. PROBLEM OVERVIEW

Measurement of e-business usage which is the measurement of the volume of its transactions is of particular interest to policy makers because of its productivity impacts. However, the ongoing challenges in this measurement are significant and include problems associated with measuring a subject that is both complex and dynamic in nature. These difficulties are exacerbated by limitations imposed by the statistical vehicles used to collect e-business usage data. While this situation is especially more problematic for businesses thinking about engaging in electronic commerce, they can gather such data from their own internal data collection methods to estimate them [2].

In [2] it is also noted that "a different situation is faced by public policy makers who are fearful that the technical characteristics, flow level, and speed of electronic commerce weaken existing policy instruments ranging from establishing a business to matters such as calculating the revenue taxes". Aside from concerns about the utility of existing policies or the need for new rules, policy makers also have a keen interest in outlining the impact that this new form of commerce will have on employment, productivity, trade, and growth. Questions such as the following summarizes such concerns: Will electronic commerce lead to a large scale substitution of new, less labor-intensive intermediaries for traditional wholesale and retail trade? Will these savings be passed onto consumers and lead to a reduction in prices? Will electronic commerce further erode the business ability to track international trade and thus weaken their ability to measure GDP and its growth? To answer these questions, policy makers need data on electronic commerce to judge the size, speed, and direction of such phenomena [1].

To be more specific, the difficulty of measurement starts from finding what constitutes electronic commerce, the speed of its growth and evolution especially when the firms conduct both electronic commerce and traditional commerce simultaneously. As discussed before and discussed in [2], quantifying the value associated with electronic commerce activities can be challenging since many of its key qualities – convenience, variety and ease of access to information – are difficult to measure.

While the supply side of the e-commerce infrastructure is currently a major drive of economic activity, many expect the demand side – the buying and selling of products over this infrastructure – to be the key source of growth in the future [2]. In terms of policy relevance, the demand side is where questions of consumer protection, taxation, payment security, parcel delivery, and the need for an updated commercial code arise. For these issues, the limited definition [1, page 172] is preferable. These policies require the separation of business-to-business transactions and those that are business-to-consumer transactions. In addition to this classification, it is useful to further separate the activities into those that involve digital products (bits or intangibles) and those that are not (atoms or tangibles) [12]. So, this leads to a situation where the traditional methods of data collection is not working anymore and new measurement/data collection methods needs to be implemented for electronic commerce.

4. NEGOTIATION METHOD FOR INTELLIGENT AGENTS

By increasing the degree (and sophistication) of automation, B2B e-Commerce will become much more dynamic, efficient, and hence more widely adopted by organizations. As indicated in [5], intelligent software agents are promising to enhance the degree of automation and sophistication of B2B e-Commerce. Software agents are encapsulated computer systems situated in some environments such as the Internet and are capable of performing flexible, autonomous actions in such environments to meet the design objectives [6]. The notion of agency can be applied to build robust architectures for automated negotiation systems within which a group of software agents communicate and autonomously make negotiation decisions on behalf of their human users. It is believed that negotiation agents can reduce human negotiation time and identify optimal or near optimal solutions from combinatorial complex negotiation spaces [6]. Negotiations are ubiquitous and conducted in various contexts such as the formation of virtual enterprises [7], marketing, establishing business contracts, managing labor disputes [9], resolving border conflicts, handling hostage crisis [5], etc. Negotiation refers to the process by which group of agents (human or software) communicate with one another in order to reach a mutually acceptable agreement on resource allocation (distribution) [11]. Given the ubiquity and importance of negotiations in various contexts, research into negotiation theories and techniques has attracted attention from multiple disciplines such as distributed artificial intelligence (DAI) [7], social psychology [8], game theory [8], operational research [10], and more recently in agent-mediated electronic commerce [6]. Despite the variety of approaches towards the study of negotiation theory, a negotiation model consists of four main elements: negotiation pro-

tocols, negotiation strategies, negotiation environments and agent environments [3].

Practical negotiation mechanisms for B2B e-commerce must be computationally efficient. Therefore, negotiation agents should be developed based on the assumption of bounded rather than perfect rationality [4]. In addition, as discussed in [5] "since individual businesses have the freedom to employ their own negotiation strategies and mechanisms to negotiate with their business partners, it implies that a practical negotiation mechanism should be based on a distributed rather than a centralized decision making model. As an agent (i.e., the representative of a business) only knows its own preferences (utility function) but not the preferences of its opponents" [5] in typical B2B negotiation settings, negotiation agents must be able to make sensible decisions based on incomplete and uncertain information. Since B2B negotiations often involve multiple parties who will exploit many issues (e.g., price, quantity, product quality) simultaneously, practical negotiation mechanisms must be able to support multilateral multi-issue negotiations. In particular, B2B relationships and interactions should be modeled as integrative negotiations [10] (i.e., the cooperative process of resolving multiple interdependent but not mutually exclusive goals). This is in stark contrast to distributive negotiation in B2C where win–lose type of negotiations are not uncommon. In short, the goal of an automated negotiation system is to search for a win–win situation efficiently given the specific preferences of the participating human negotiators.

To address the issue, it seems that multi agent system (MAS) is a good candidate. MAS is system which is composed of several intelligent agents (IA) in interaction with each other [4]. Such systems are used when it cannot be managed via a single IA or it is difficult to control; for instance, online business and designing social structures. In [4], the important characteristics of agents of MAS are listed as:

1. Choice: the agents are at least partially independent.
2. Local view: no agent has a complete view of the system. Usually the system is too complicated for a single agent to make the most of the system from one view point.
3. No centrality: no central agent controls the other agents

In other words, MAS is composed of several independent and synchronic agents which cooperate with each other to reach a common goal. IAS (intelligent agents) existing in such a system are connected with each other via message communication. The language of these messages is ACL .One of these languages is ebXML as discussed before. The messages that agents send to each other have different meanings such as comparison, commands, or structure. On the other hand, web services are the available software elements for the agents which will give useful information to users. So they are useful for IAS.

Finally, these IASs will be more powerful in a complicated situation dealing with bulk of information to help users. It also can take the responsibility of collecting information, buyers, sellers, and preparation.

There are different rules on the activity of agents which must be observed when something occurs during the time the agents are in touch with one another, for instance, when a suggestion is offered or an answering time ends with no reply. These rules are divided into 3 categories [7]:

Suggestion rules: the function of this group is to control and manage the suggestions and requests expressed by other agents .It should be clear if these suggestions are in accordance with the main mechanism of the communication among agents both conceptually and structurally and the requests are analyzed both structurally and conceptually. For example, here it will be decided if the request is allowed to be sent or not.

Information rules: in each agent, the receiver of the information is necessarily a receiver of data whose job is to analyze them .These rules controls the condition within which this information is made. Indubitably, this is necessary in the process of communication. The changes in the situation of requests and communication are done within the limits of these rules.

Clarification rules: the final goal of each negotiation among the intelligent agents is to complete one or several business among the parties of the negotiation. The last group of rules registers and calculates this business, totally it controls the negotiations.

5. RELATED WORKS

Using business intelligent agents to connect the B2B markets has been experienced in different projects and structures. However, using this method to measure the bulk of B2B transactions has not been reported yet. Only in one case, the writer in paper[4] offered a model in order for the governments to have more control on business markets. In that model, one agent had the responsibility to run negotiation among different business and sent their information to the central server.

6. SOLUTION

According to what was mentioned above, business intelligent agents can nicely connect the businesses to each other for data communication and business. the intelligent agents (IAs) can negotiate with each other through the standard of ebXML and send the requested data to the other agent .In one business cooperation, the agents should have the same ebXML dictionary to connect and negotiate with one understandable language .In the model offered in this paper, One main repository has been considered in such a way that when the businesses enroll as a member in it, they can not only be allowed to connect the net and its software ,but they also can receive the registry of this repository and use while sending and receiving information. The structure of this repository is in a way that the information of a registered business after analyzing is considered as an allowed business and is included in the list of IAS (Intelligent Agent System) and portable robots in order to be sent to website of the business and receive the necessary information.

Each business after membership is allowed to install a XML web service in its website. This web service sends the

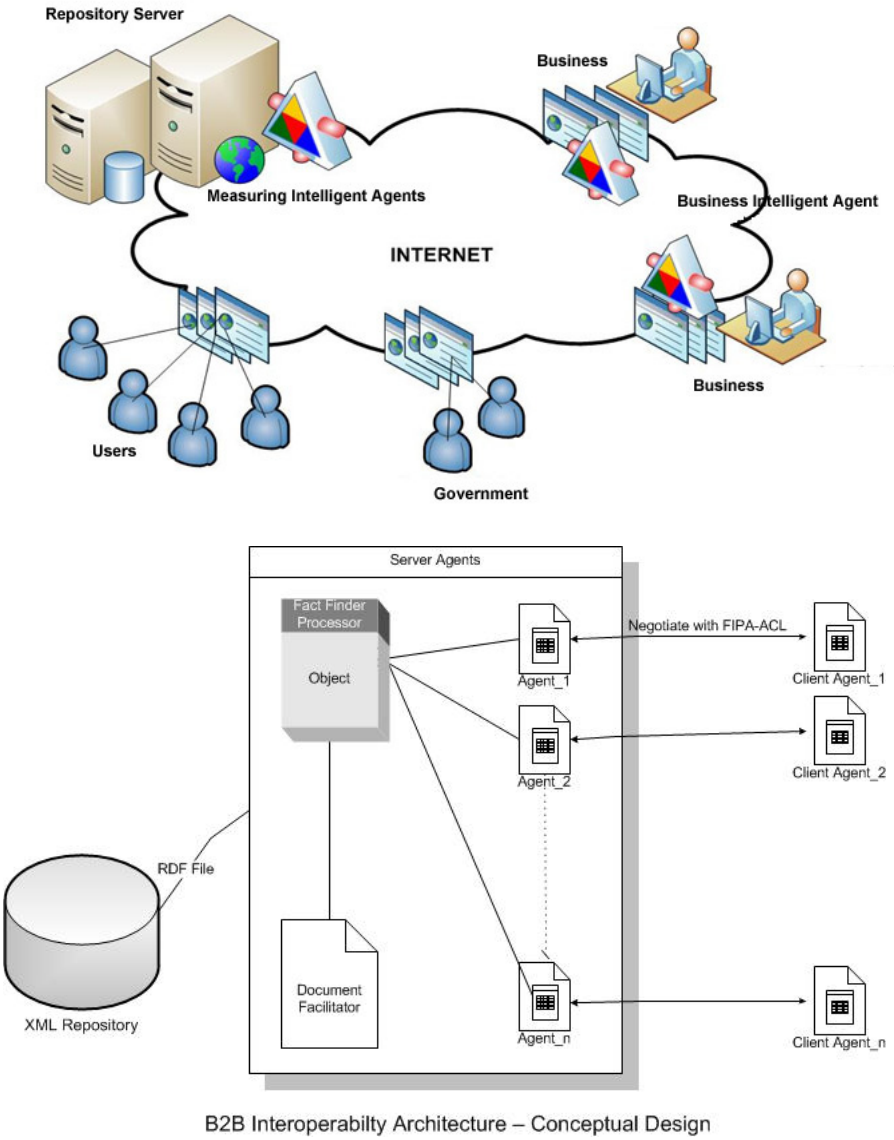

Figure 1 Conceptual Acrchitecture.

needed and agreed upon information to the repository intelligent agent.

All the communications in this system is through ebXML messages and all the messages are processed and sent via IAS. The technology used in this in the IAS of this system is web service which is used in repository server and also is installed in that business too. However, the function of the web service at the business side and server is different. The responsibility of the server at the business side is to send the information of the amount of the business transaction to other businesses. After each transaction between the business and the business partner, the related web service is called and sends the needed information to measure the bulk of the transaction to sever. This information includes the following items: 1. Company properties; 2. Time; 3. International code of the goods or service; 4. The number of transaction; 5. The price of transaction; 6. Tax; 7. Other reductions; 8. Way of payment; 9. Way of sending.

After sending the above mentioned information to the server, the IA of the server receives and registers this information at the repository database. The IA of the server after doing all this will send a message to the web service of the business announcing the successful registering of sent information. If the both businesses who are doing a business are member of the repository, the IA of the server will recognize the sent information from the seller as reliable, and registers it and doesn't consider the buyers information.

The software designed for the repository server contains the following capabilities:

1. The possibility of instant observing of the bulk of transaction: in this section there is the possibility to observe the amount of the business transaction in a certain time period to help make better decisions.

2. Observing the amount of the transaction of a certain goods: if the policy makers intend to know the amount

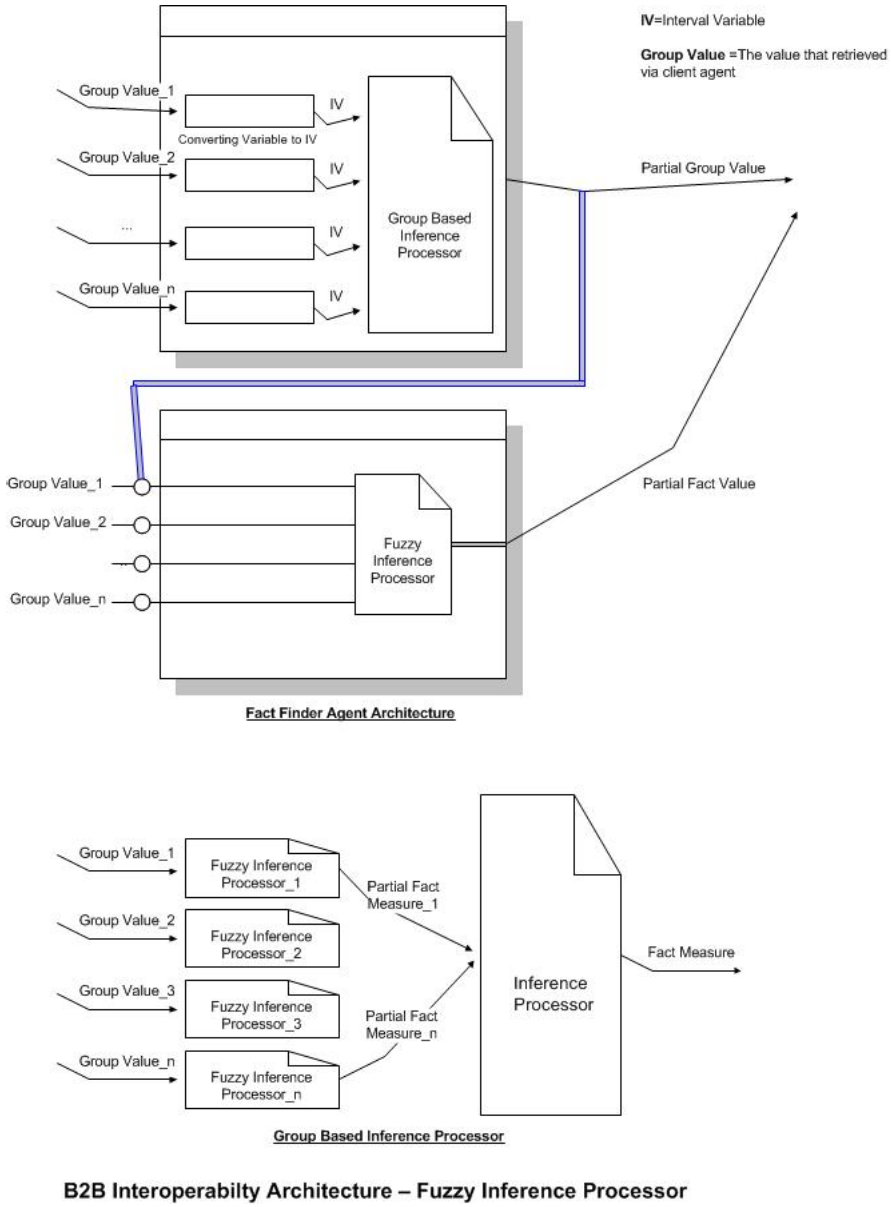

Figure 2 B2BFIS Model.

of transaction of a certain part, the information existing in the repository can be of great help.

6.1 The Architecture

The architecture designed for this model is shown in figure 1. The main sever which is considered as repository sends the IAS to the websites of the businesses. The website of the business is on the other side, which is connected to the server by the help of translator and transmitter IA. A XML web service is also installed on each of the websites of the businesses to extract the requested information of transmitter IA.

In this model, we have two groups of agents. First group is comprised of server agents and the second one of the client agents. Once users download the client agent and register the specifications, can connect to the agents of server. The server agent after connecting to the client agent sends the query to the client agent, the client agent, likewise, gets answers from the human user and will send the replies according to the categories specified in the HTML format to its corresponding server agent.

The server agent in this phase will perform the following steps:

1. Selecting the ontology of received structure conversion from HTML to XML.

2. Performing the process of HTML conversion to XML in the format of RDF.

3. Transmitting the gained RDF structure to Fact Finder Agent.

4. Getting the responses based on the accuracy rate of offered replies and transmitting to the server agent.

5. Selecting the group of subsequent questions by server agent and transmitting it to the client agent.

Communication Language between server and client agents in order to negotiate is LARKS language. LARKS is a language to define the agent capabilities that enables the advertising, application and compliance of capabilities for the agents. One of the advantages of this Communication language is the content portability among the agents with the command.

LARKS language includes the following:

- Message content or describing the agent demand
- Types of data used in the message body
- Input and output parameters and their restrictions which will be in the form of " Horn clauses"
- Text description regarding the service or application (TextDescription)
- concept description (ConcDescription)

After receiving the HTML structure of responses and converting them to RDF, the server-agent to discover the given responses reality will send the RDF file to the processing-agent which discovers the reality.

This operating procedure has been introduced in Figure 2. The architecture of questions initially was separated to the related several groups, in a way that questions in each group were the complements of the other groups and this can cause by using Fuzzy Logic to derive the weight of reality hidden in any of the responses [13].

According to this method, any group of the responses will be sent to a Fuzzy-Logic motor that in this research was used of MAMDANI motor and the received output that is the weight of that group will be sent Deduction-Processor [13,14].

The output obtained from every group in the format of input again will be sent to Deduction–Motor of MAMDANI. Some partial amount obtained with the operator with each amount of the previous phase will become XOR and again will enter the Deduction-Motor of MAMDANI and the obtained output is the weight of responses reality.

The weight of responses' reality will be sent to the server and the server agent also according to the classification which was done previously will send the next group of the questions or will decide to send again the previous questions. This action will be repeated for the total groups to be received the responses of total questions.

7. CONCLUSION AND FURTHER WORK

Measuring the amount of electronic business is as important as the electronic business itself. Therefore, having the data and updated and exact information of the transactions seems to be necessary. The high amount of financial transactions in B2B compared to B2C and the difficulty in measuring these transactions via existing traditional methods are among the challenges the investigators face. Thus, it seems that for this problem we should use a method using the problem itself. In this paper, a model is suggested which can greatly help collect online and exact amount of B2B transactions. To widely execute this model and force the businesses to register at the repository server, we can use different controlling methods such as: tax declaration time, receiving digital signature, receiving operation license, etc. Another advantage of this model is that the information of the businesses can be changed easily compared to traditional methods which were very expensive.

We also intend to offer a model to measure the amount of business transaction of B2C which has its own challenges due to its small size (or other features).

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